Statistical Learning and Data Analysis Module B

Principal Component Analysis

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Statistics, tèchnè-loghìa, analysis of data

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A thought

- ***«entia non sunt multiplicanda praeter necessitatem»,* G. Occam (XIV sec.)**
- «Things should not be multiplied beyond what is required»

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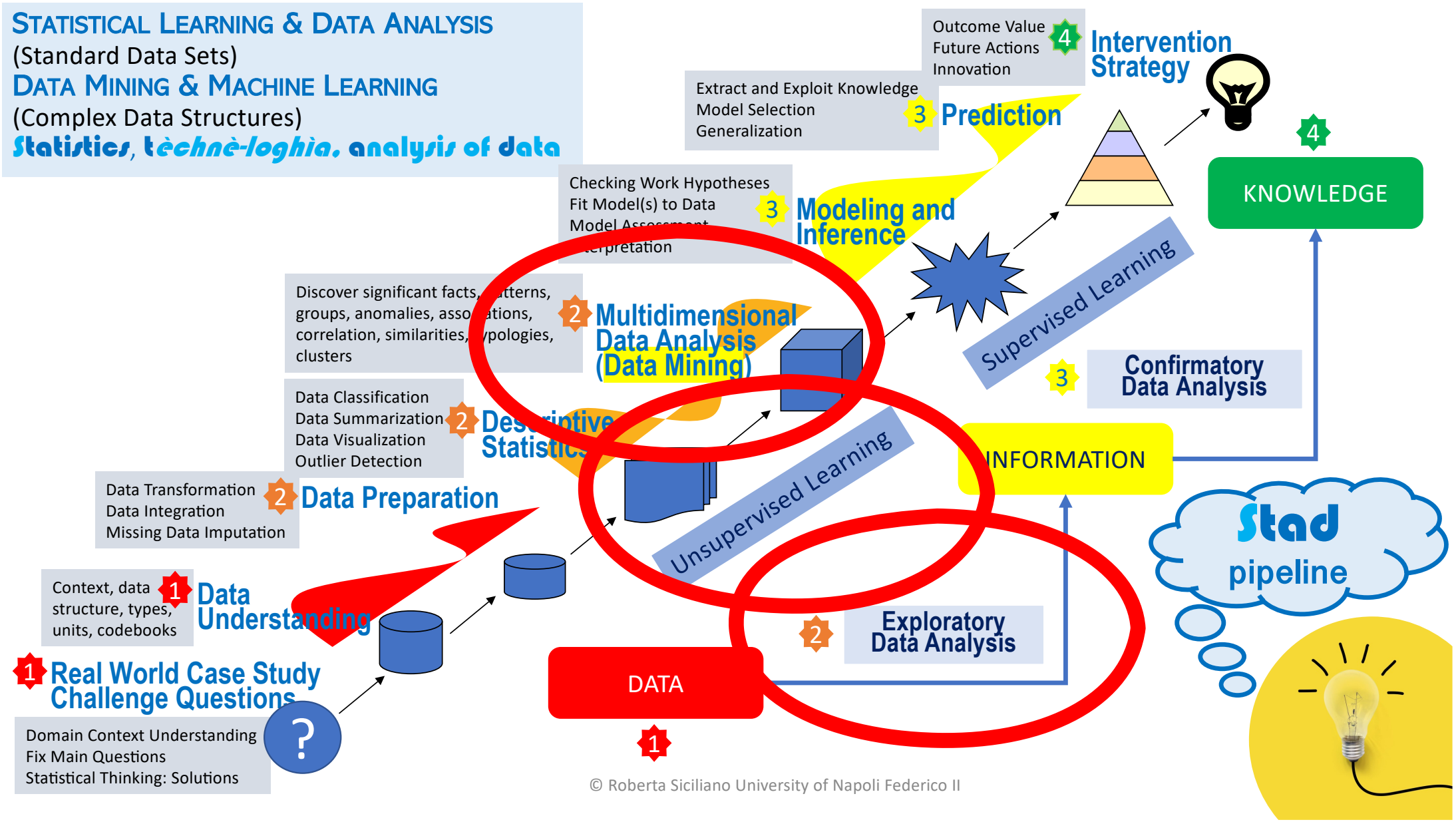
STATISTICAL LEARNING & DATA ANALYSIS
(Standard Data Sets)
DATA MINING & MACHINE LEARNING
(Complex Data Structures)
Statistics, tèchnè-loghìa, analysis of data
Outcome Value
Future Actions
Innovation
4
Intervention Strategy
Extract and Exploit Knowledge
Model Selection
Generalization
3
Prediction
4
KNOWLEDGE
Checking Work Hypotheses
Fit Model(s) to Data
3
Modeling and Inference
Supervised Learning
Discover significant facts, patterns, groups, anomalies, associations, correlation, similarities, typologies, clusters
2
Multidimensional Data Analysis (Data Mining)
3
Confirmatory Data Analysis
Data Classification
Data Summarization
Data Visualization
Outlier Detection
2
Descriptive Statistics
INFORMATION
Unsupervised Learning
Data Transformation
Data Integration
Missing Data Imputation
2
Data Preparation
Stad pipeline
Context, data structure, types, units, codebooks
1
Data Understanding
2
Exploratory Data Analysis
1
Real World Case Study Challenge Questions
DATA
Domain Context Understanding
Fix Main Questions
Statistical Thinking: Solutions
?
1
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Factorial Methods

Principal Component Analysis

Correspondence Analysis

Multiple Correspondence Analysis

Gifi System for Non Linear Multivariate Analysis

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Historical flashback

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Multivariate or Multidimensional Data Analysis?

Confirmative Approach (*Classical Statistics*, Hotelling, anni '30):

- **Starting point**: extension of univariate and bivariate probabilistic models to more than two variables (multivariate statistics).
- **Objective**: construction of explanatory models for the reconstruction of multivariate data that allow to confirm the hypotheses formulated on the distribution of the variables.

Exploratory Approach (*Analyse des données*, Benzecrì, anni '70):

- **Starting point**: broad system of elementary data and implausibility of probabilistic assumptions.
- **Objective**: to geometrically describe the complexity of the phenomenon under study.

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Contents

Geometrical View of the Data Matrix

Principal Component Analysis (PCA)

Singular Value Decomposition (SVD)

PCA Fitting Criterion

PCA as Reduced-Rank Model

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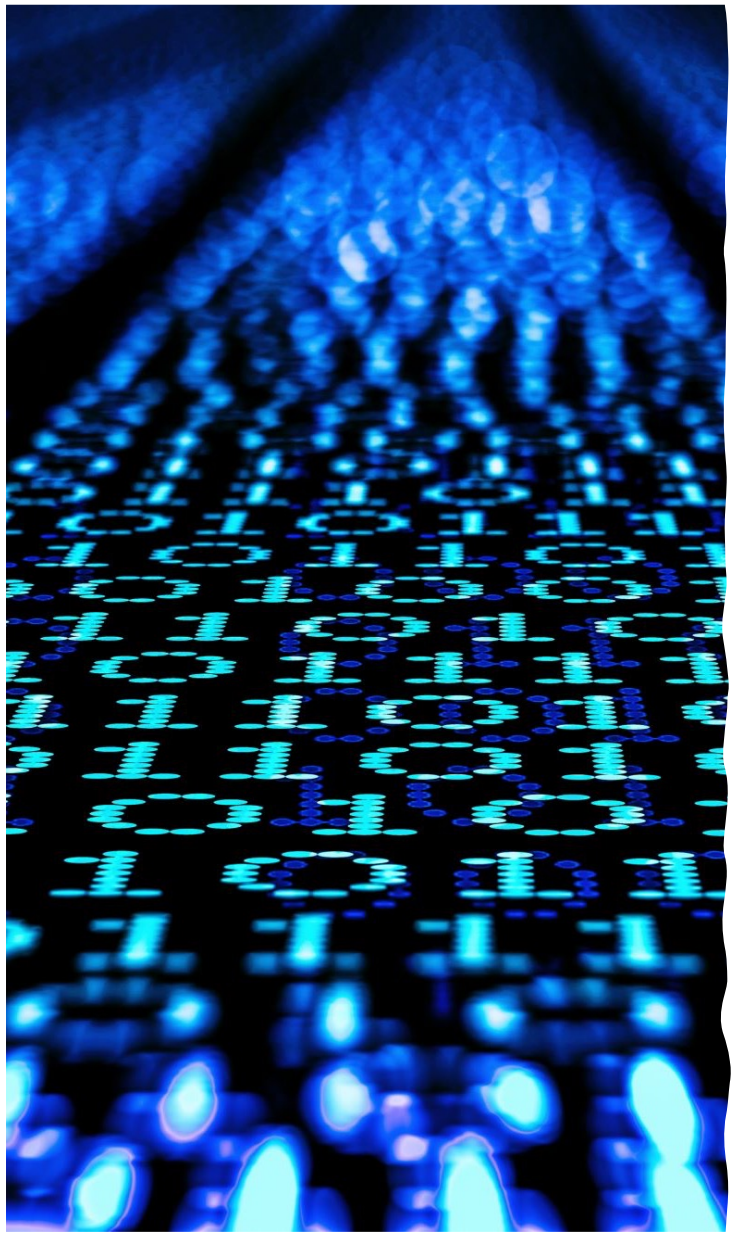

Data

A multidimensional data matrix X includes the measurements of p variables, also known as *features*, on n units, also known as *observations*.

- Each row of X is a multidimensional data, namely the i-th record or row vector contains p features measurements on the i-th observation:

$x_i' = (x_{i1}, \dots, x_{ip})'$ for $i = 1, \dots, n$.

- Each column refers to the unitary distribution of a variable, namely the j-th column or column vector contains the measurement of the j-th feature on all observations:

$x_j = (x_{1j}, \dots, x_{nj})$ for $j = 1, \dots, p$.

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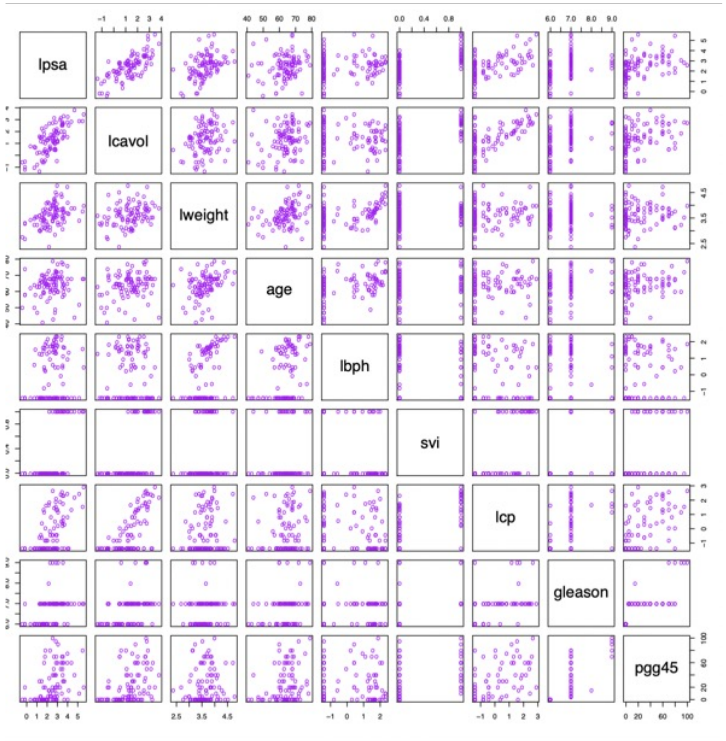
Linear Correlation

- How to visualize linear correlation between each couple of features?
- Scatterplot matrix visualizes the correlation of each pair of features for a total number of $\binom{p}{2} = \frac{p(p-1)}{2}$ scatterplots.
- Say for $p = 8$, there are 28 plots to analyze!



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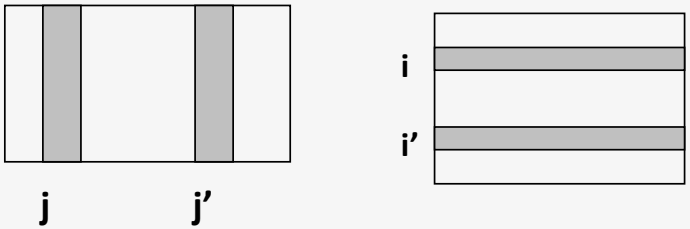

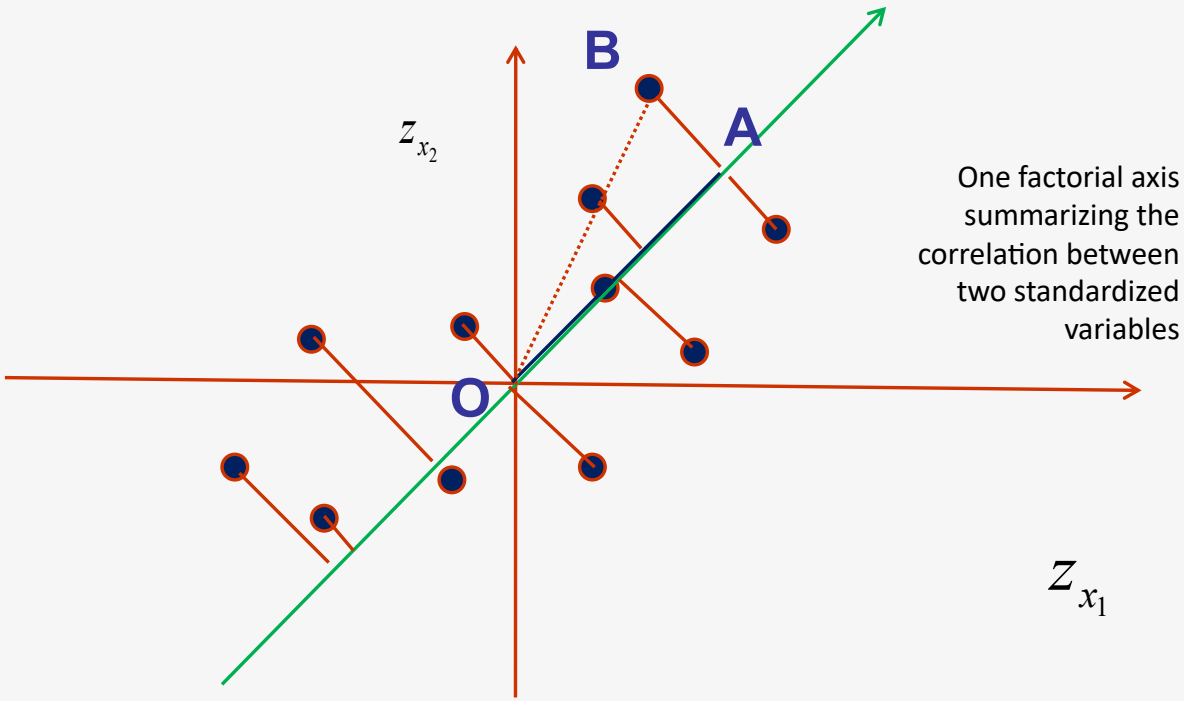

One factorial axis summarizing the correlation between two standardized variables

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Geometrical View of the Data Matrix

Row vectors of **X** are	• **Unit-points** in $\Re^p$ spanned by the variables
Columns vectors of **X** are	• **Variable-points** in $\Re^n$ spanned by the units

Principal Component Analysis

- Let X be the $n \times p$ data matrix including the measurements of p numerical features on n observations.
- Goal of PCA is **to reduce the dimensionality of the data** set by *eliminating the redundancy of information resulting from p highly correlated variables* and by *replacing* the latter with *a smaller number M (with $M < p$) of new (latent) variables that are*
  - *not correlated with each other,*
  - *linearly linked to the starting variables,*
  - *ordered with respect to the percentage of variability present in the original data.*
- If it is not possible to fully represent the original variability with fewer M variables, **PCA is limited to representing most of this variability with a smaller number of variables.**

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The First Principal Component (PC)

- Let assume that the $n \times p$ data matrix X is centered so that the features have mean zero (the column means in X are zero).
- The First Principal Component (PC1) is the normalized linear combination of the original features:

$$Z_1 = u_{11}X_1 + \ldots + u_{p1}\,X_p$$

that has the largest variance $var(Z_1) = \frac{1}{n}\sum_{i=1}^{n} z_{i1}^2$.

- By normalized, we mean that $\sum_{j=1}^{p} u_{j1}^2 = 1$. Otherwise setting these elements to be arbitrarily large in absolute value could result in an arbitrarily large variance.
- PC1 can also be considered the closest correlate to all observed features.
- PC1 is a single-variable summary of what the original features most have in common.

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Optimization problem

- In **matrix notation**, the first PC is defined as $z_1 = Xu_1$, where
  - u_1 is a p-dimensional column vector with the elements $(u_{11}, \dots, u_{p1})$ called **PC factor loadings,**
  - z_1 is a n-dimensional column vector which elements $(z_{11}, \dots, z_{n1})$ called **PC scores.**
- In **scalar form** the i-th unit-point has as score of the first PC given by

$$z_{i1} = u_{11}x_{i1} + \dots + u_{p1}\, x_{ip}$$

- The factor loadings are found by maximizing the sample variance of the n values z_{i1}, for $i = 1, \dots, n$, given by $var(Z_1) = \frac{1}{n}\sum_{i=1}^{n} z_{i1}^2$.
- This means to solve the optimization problem:

$$\max_{u_{11},\dots,u_{p1}} \left\{\frac{1}{n}\sum_{i=1}^{n}\left(\sum_{j=1}^{p} u_{j1}x_{ij}\right)^2\right\} subject\ to \sum_{j=1}^{p} u_{j1}^2 = 1.$$

- *This problem can be solved via an eigen decomposition, a standard technique in linear algebra*.

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The second PC

- The Second Principal Component is the normalized linear combination of the original features:
$$Z_2 = u_{12}X_1 + \ldots + u_{p2}\,X_p$$
- that has the largest variance $var(Z_2) = \frac{1}{n}\sum_{i=1}^{n} z_{i2}^2$, with the constraints $\sum_{j=1}^{p} u_{j2}^2 = 1$ and $\sum_{j=1}^{p} u_{j1}u_{j2} = 0$.
- Constraining Z_2 to be uncorrelated with Z_1 is equivalent to constraining the direction of the vector u_2 to be **orthogonal (perpendicular)** to the direction the vector u_1.

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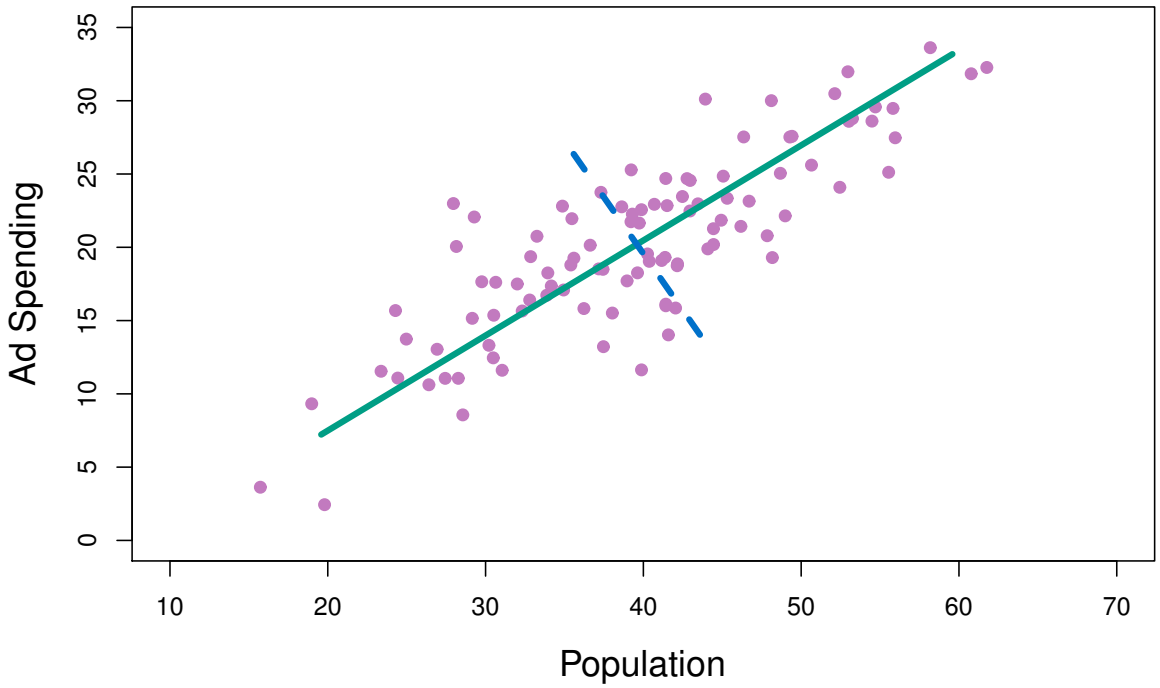
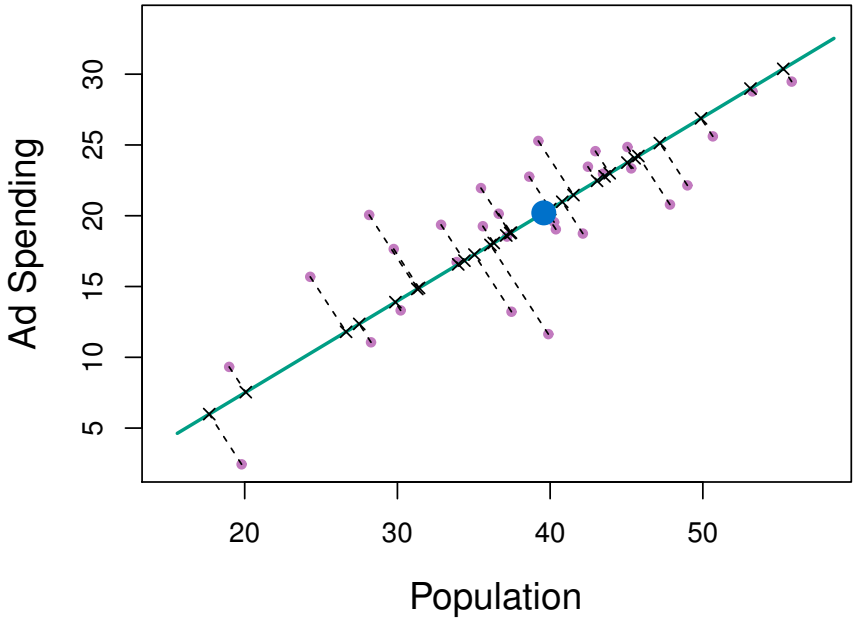
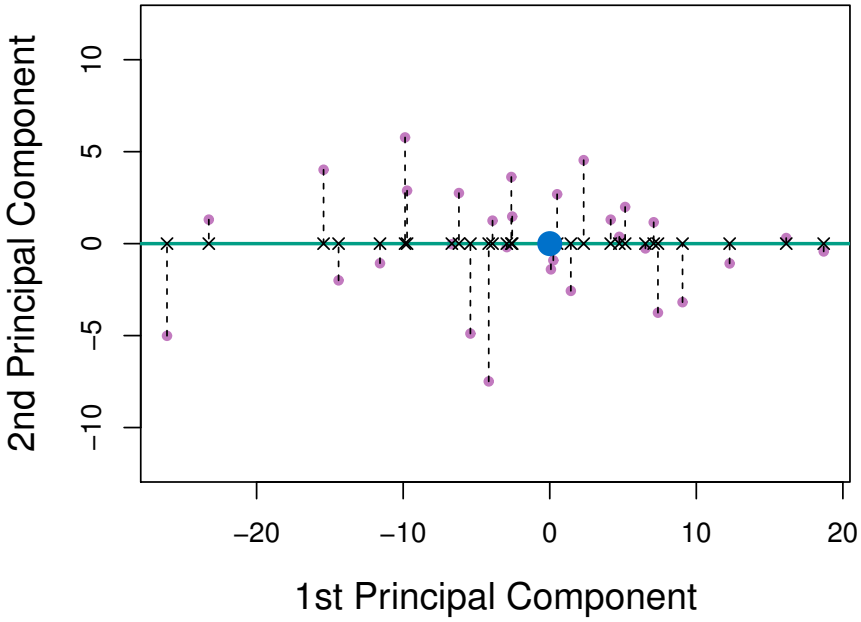
PCA on Population size and Ad Spending for 100 cities

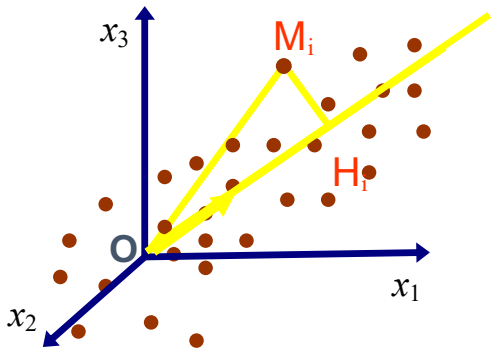


Most of the variance captured by the first PC





An example



Geometrical view

u=[1 0 1]

$||u||^2 = 1^2 + 0^2 + 1^2 = 2$

Normalized vector

$$\mathbf{u} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Projection of a point on u

M_i=[3 2 5]

$$OH_i = \begin{bmatrix} 3 & 2 & 5 \end{bmatrix} \times \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = \frac{3}{\sqrt{2}} + 0 + \frac{5}{\sqrt{2}} = 5{,}65$$

$$\sum_{i=1}^{n} (OH_t)^2 = (\mathbf{Xu})' \mathbf{Xu} = \mathbf{u}'\mathbf{X}'\mathbf{Xu}$$

$$\begin{cases} \max_{(\mathbf{u})} [\mathbf{u}'\mathbf{X}'\mathbf{Xu}] \\ \mathbf{u}'\mathbf{u} = 1 \end{cases}$$

Analytical method

$$\sum_{i=1}^{n} (M_t H_t)^2 = \min \quad \Leftrightarrow \quad \sum_{i=1}^{n} (OH_t)^2 = \max$$

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Singular Value Decomposition (SVD)

- If $\mathbf{X}$ is a rectangular matrix of rank $r = \min(n, p)$ (usually $p < n$), the SVD of $\mathbf{X}$ is given by
$$\mathbf{X} = \mathbf{V}\boldsymbol{\Gamma}\mathbf{U}'$$
- where $\mathbf{U}$ and $\mathbf{V}$ are matrices which columns are orthonormal
$$\mathbf{U}'\mathbf{U} = \mathbf{I}, \mathbf{V}'\mathbf{V} = \mathbf{I}$$
- and $\boldsymbol{\Gamma}$ is diagonal with positive elements in nonincreasing order
$$\boldsymbol{\Gamma} = diag(\tau_k)$$

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Spectral Decompositions

- The r columns of $\mathbf{V}$ are the *left* eigenvectors of $\mathbf{XX}'$ since $\mathbf{XX}' = \mathbf{V\Gamma U}'\mathbf{U\Gamma}'\mathbf{V}' = \mathbf{V\Lambda V}'$ being $\mathbf{U}'\mathbf{U} = \mathbf{I}$ and $\mathbf{\Gamma\Gamma}' = \mathbf{\Lambda}$ (the matrix with the eigenvalues).

- The r columns of $\mathbf{U}$ are the *right* eigenvectors of $\mathbf{X}'\mathbf{X}$ since $\mathbf{X}'\mathbf{X} = \mathbf{U\Gamma}'\mathbf{V}'\mathbf{V\Gamma U}' = \mathbf{U\Lambda U}'$ being $\mathbf{V}'\mathbf{V} = \mathbf{I}$ and $\mathbf{\Gamma\Gamma}' = \mathbf{\Lambda}$.

- The eigenvalues matrix $\mathbf{\Lambda} = \mathbf{\Gamma}'\mathbf{\Gamma} = diag(\lambda_k)$ is such that $\lambda_k = \tau_k^2$.

- $tr(\mathbf{X}'\mathbf{X}) = \sum_{k=1}^{p} \lambda_k$. The p-dimensional space defined by the factorial axes exactly reconstructs the variability of the cloud of points in the original space R^p

- Being $\mathbf{X}'\mathbf{X}$ a symmetric and positive definite matrix the eigenvalues of $\mathbf{XX}'$are coincident to the eigenvalues of $\mathbf{X}'\mathbf{X}$.

- Transition formula: $\mathbf{U} = \mathbf{X}'\mathbf{V\Gamma}^{-1}, \mathbf{V} = \mathbf{XU\Gamma}^{-1}$.

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$$\mathbf{X} = \mathbf{V\Gamma U^T}$$

$$X = \begin{bmatrix} 1.1547 & -1.1547 \\ -1.0774 & 0.0774 \\ -0.0774 & 1.0774 \end{bmatrix} = \begin{bmatrix} 0.8165 & 0 \\ -0.4082 & -0.7071 \\ -0.4082 & 0.7071 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.7071 & 0.7071 \\ -0.7071 & 0.7071 \end{bmatrix} = \begin{bmatrix} 1.1547 & -1.1547 \\ -1.0774 & 0.0774 \\ -0.0774 & 1.0774 \end{bmatrix}$$

$$XX^T = V\Lambda V^T$$

$$XX^T = \begin{bmatrix} 0.8165 & 0 \\ -0.4082 & -0.7071 \\ -0.4082 & 0.7071 \end{bmatrix} \begin{bmatrix} 2^2 & 0 \\ 0 & 1^2 \end{bmatrix} \begin{bmatrix} 0.8165 & -0.4082 & -0.4082 \\ 0 & -0.7071 & 0.7071 \end{bmatrix} = \begin{bmatrix} 2.6667 & -1.3333 & -1.3333 \\ -1.3333 & 1.1667 & 0.1667 \\ -1.3333 & 0.1667 & 1.1667 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 0.7071 & 0.7071 \\ -0.7071 & 0.7071 \end{bmatrix} \begin{bmatrix} 2^2 & 0 \\ 0 & 1^2 \end{bmatrix} \begin{bmatrix} 0.7071 & -0.7071 \\ 0.7071 & 0.7071 \end{bmatrix} = \begin{bmatrix} 2.5 & -1.5 \\ -1.5 & 2.5 \end{bmatrix}$$

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Euclidean distance between two variable-points

$$d^2(j,j') = \sum_{i=1}^{n} \left(x_{ij} - x_{ij'}\right)^2 =$$

$$= \sum_i x_{ij}^2 + \sum_i x_{ij'}^2 - 2\sum_i x_{ij} x_{ij'} =$$

$$= \sum_i \left(\frac{x_{ij} - \mu_j}{\sigma_j \sqrt{n}}\right)^2 + \sum_i \left(\frac{x_{ij'} - \mu_{j'}}{\sigma_{j'} \sqrt{n}}\right)^2 - 2\sum_i \left(\frac{x_{ij} - \mu_j}{\sigma_j \sqrt{n}}\right)\left(\frac{x_{ij'} - \mu_{j'}}{\sigma_{j'} \sqrt{n}}\right) =$$

$$= \frac{1}{\sigma_j^2}\sigma_j^2 + \frac{1}{\sigma_{j'}^2}\sigma_{j'}^2 - 2\rho_{jj'} = 2\left(1 - \rho_{jj'}\right)$$

- Study the distances and proximities is equivalent to study the linear correlation.

SVD for PCA

PCA using Centered Data

- Searching for eigenvectors and eigenvalues of the Covariance Matrix $X'X = S$

PCA using Standardized Data

- Searching for eigenvectors and eigenvalues of the Correlation Matrix $X'X = R$

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The role of the eigenvalues

- **Explained variance:**
  - The relationship is valid: $tr(\mathbf{X}'\mathbf{X}) = \sum_{k=1}^{p} \lambda_k$
  - The p-dimensional space defined by the factorial axes exactly reconstructs the variability of the cloud of points in the original space R^p
  - λ_k represents the variance explained by the k -th principal component.
- **Selection of the number of factorial axes**
  - Variability explained
    - a minimum threshold of explained variability is set
  - Eigenvalue-one (for standardized variables)
    - Since the variables have unit variance, only the eigenvalues greater than one are chosen
  - Scree-Test
    - the factors whose eigenvalues precede the maximum jump of explained variability are considered

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Example: Food Consumption in European Countries

Country	Cereals	Rice	Potatoes	Sugar	Vegetables	Wine	Meat	Milk	Butter	Eggs
Belgium	72.2	4.2	98.8	40.4	103.2	20.9	102	80	7.7	14.2
Denmark	70.5	2.2	57	39.5	50	22	105.8	145.2	4.1	14.3
Germany	71.3	2.3	74.1	37.1	83.1	22.8	97.2	90.7	6.9	14.8
Greece	109.8	5.4	90	30	229.5	25.3	77.1	63.1	0.9	11.3
Spain	71.4	5.8	107.8	26.8	191.7	43	102.1	98.4	0.6	15.3
France	73	4.3	78.2	34.1	95	64.5	110.5	98.9	8.9	15
Ireland	93.4	3.2	151.5	34.8	55	3.9	105	185.9	3.4	11.4
Italy	110.2	4.8	38.6	27.9	181.9	61.6	88	65	2.4	11.1
Netherlands	54.6	5	86.7	39.7	99	14	89.4	136.2	5.4	10.7
Portugal	86	5.7	106.6	29.4	100	57	75.5	96	1.5	7.7
UK	74.3	4.5	94.1	39.8	60	10.4	74.4	129.3	3.2	10.8
Austria	68.7	4.2	62.6	37.1	81.9	34.3	93.4	121.3	4.3	13.4
Finland	70.1	5.4	61.6	35.7	52.6	10.2	65	208.4	5.8	10.9
Iceland	79.7	1.9	50.2	54.9	50	6.2	71.7	205.6	4.6	11.3
Norway	76.9	3.5	73.3	7.3	48.3	6.6	54.9	176.5	2.1	11.3
Sweden	69.3	4.3	70	37.5	48.5	12.3	60.5	154.1	5.7	12.9

Data Matrix: 16 European Countries, 10 variables

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Descriptive Statistics

EFFETTIVI TOTALI : 16 PESO TOTALE : 16.00

IDEN - ETICH.	EFF.	PESO	MEDIA	SQM	MIN	MAX
CERE - cereali	16	16.00	78.21	14.45	54.60	110.20
RISO - riso	16	16.00	4.17	1.20	1.90	5.80
PATA - patate	16	16.00	81.31	26.48	38.60	151.50
ZUCC - zucchero	16	16.00	36.38	6.40	26.80	54.90
VERD - verdure	16	16.00	95.61	55.01	48.30	229.50
VINO - vino	16	16.00	25.94	19.69	3.90	64.50
CARN - carne	16	16.00	85.78	17.05	54.90	110.50
LATT - latte	16	16.00	128.41	46.03	63.10	208.40
BURR - burro	16	16.00	4.22	2.35	0.60	8.90
UOVA - uova	16	16.00	12.27	2.01	7.70	15.30

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Correlation matrix

	Cereals	Rice	Potatoes	Sugar	Vegetables	Wine	Meat	Milk	Butter	Eggs
Cere	1,00									
Riso	0,13	1,00								
Pata	0,06	0,23	1,00							
Zucc	-0,41	-0,69	-0,28	1,00						
Verd	0,56	0,57	0,07	-0,64	1,00					
Vino	0,29	0,42	-0,13	-0,62	0,54	1,00				
Carn	-0,07	-0,15	0,29	-0,19	0,22	0,39	1,00			
Latt	-0,34	-0,39	-0,04	0,58	-0,75	-0,69	-0,41	1,00		
Burr	-0,52	-0,34	-0,19	0,43	-0,46	-0,06	0,29	0,10	1,00	
Uova	-0,34	-0,31	-0,10	0,02	0,07	0,11	0,60	-0,22	0,45	1,00

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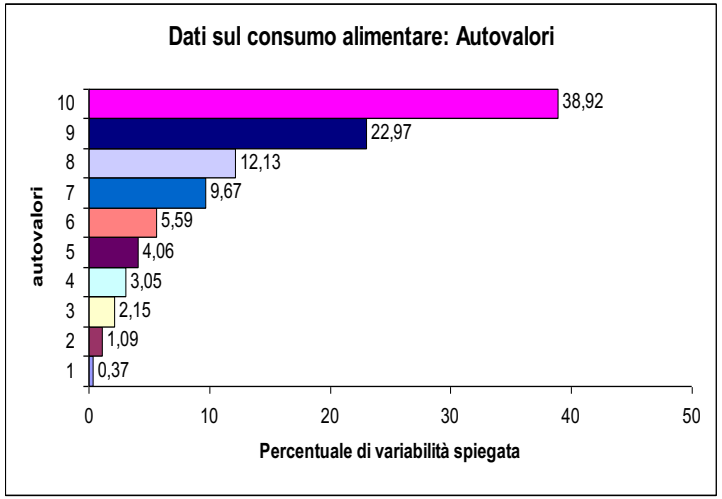
Food Consumption Matrix

Numero	Autovalore	Percentuale di inerzia spiegata	Percentuale cumulata di inerzia spiegata
1	3,8920	38,92	38,92
2	2,2966	22,97	61,89
3	1,2126	12,13	74,01
4	0,9672	9,67	83,68
5	0,5595	5,59	89,28
6	0,4064	4,06	93,34
7	0,3051	3,05	96,39
8	0,2148	2,15	98,54
9	0,1091	1,09	99,63
10	0,0367	0,37	100,00



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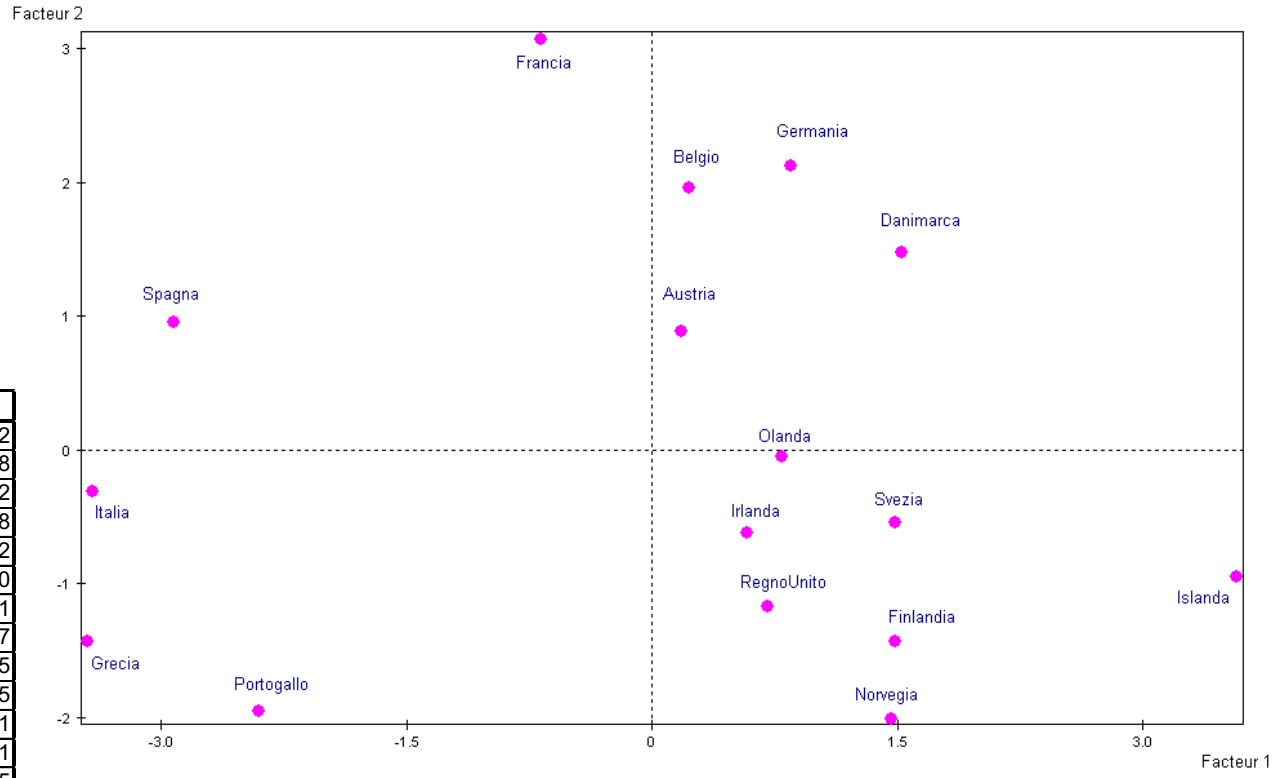
First factorial plan for units - representation

Nazione	Asse 1	Asse 2	Asse 3	Asse 4	Asse 5
Belgio	0,23	1,96	0,62	-0,10	0,12
Danimarca	1,52	1,48	-0,43	1,00	-0,28
Germania	0,85	2,12	-0,30	0,64	-0,12
Grecia	-3,45	-1,43	-0,32	1,14	-0,78
Spagna	-2,92	0,95	1,23	-0,39	-1,72
Francia	-0,68	3,08	-0,32	-0,77	1,10
Irlanda	0,58	-0,62	2,93	1,78	0,81
Italia	-3,42	-0,31	-2,27	0,75	0,57
Olanda	0,79	-0,04	0,67	-1,36	-0,05
Portogallo	-2,40	-1,96	0,42	-0,83	1,55
RegnoUnito	0,70	-1,17	0,54	-0,25	-0,01
Austria	0,18	0,89	-0,48	-0,42	-0,21
Finlandia	1,48	-1,43	-0,26	-1,63	-0,05
Islanda	3,58	-0,95	-1,35	1,36	0,14
Norvegia	1,46	-2,02	-0,29	0,07	-0,63
Svezia	1,49	-0,54	-0,39	-0,99	-0,46



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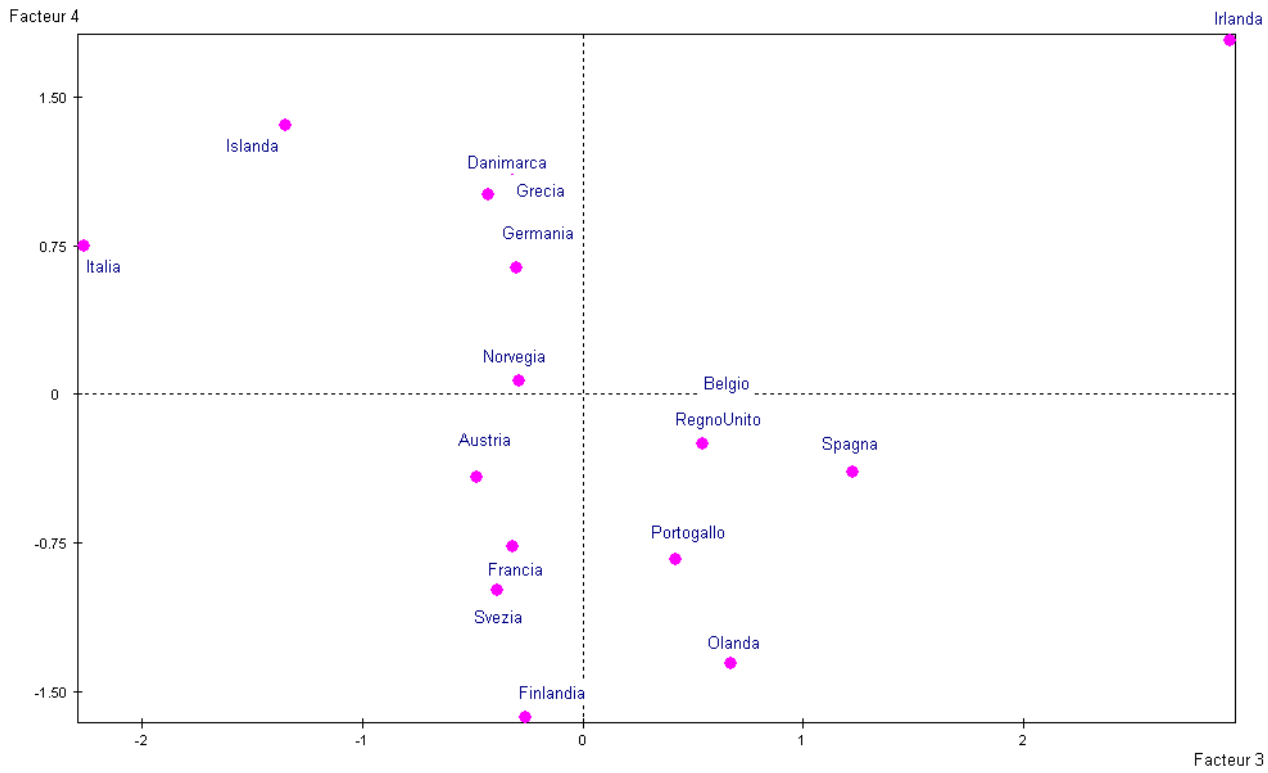
Second factorial plan for units - representation

Nazione	Asse 1	Asse 2	Asse 3	Asse 4	Asse 5
Belgio	0,23	1,96	0,62	-0,10	0,12
Danimarca	1,52	1,48	-0,43	1,00	-0,28
Germania	0,85	2,12	-0,30	0,64	-0,12
Grecia	-3,45	-1,43	-0,32	1,14	-0,78
Spagna	-2,92	0,95	1,23	-0,39	-1,72
Francia	-0,68	3,08	-0,32	-0,77	1,10
Irlanda	0,58	-0,62	2,93	1,78	0,81
Italia	-3,42	-0,31	-2,27	0,75	0,57
Olanda	0,79	-0,04	0,67	-1,36	-0,05
Portogallo	-2,40	-1,96	0,42	-0,83	1,55
RegnoUnito	0,70	-1,17	0,54	-0,25	-0,01
Austria	0,18	0,89	-0,48	-0,42	-0,21
Finlandia	1,48	-1,43	-0,26	-1,63	-0,05
Islanda	3,58	-0,95	-1,35	1,36	0,14
Norvegia	1,46	-2,02	-0,29	0,07	-0,63
Svezia	1,49	-0,54	-0,39	-0,99	-0,46



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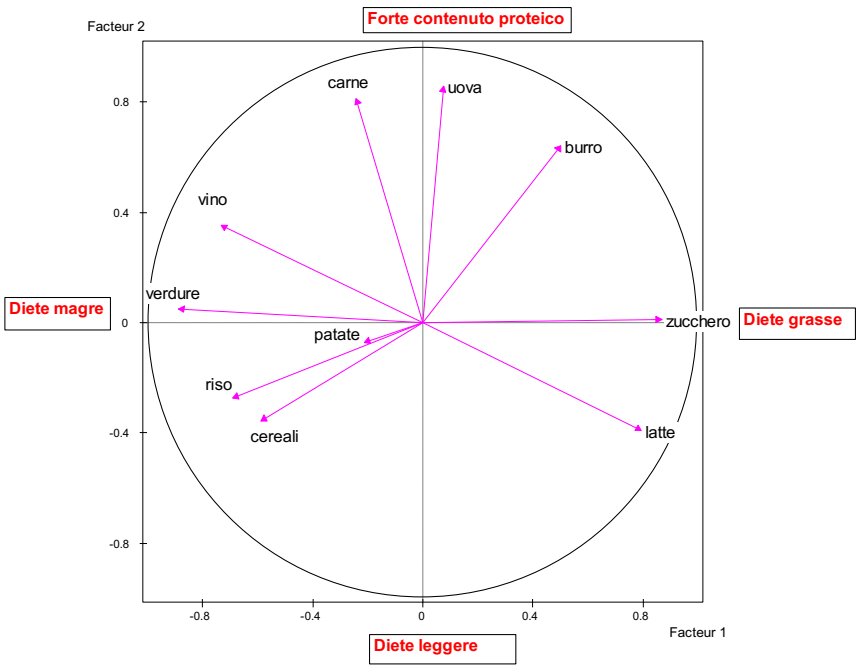
First factorial plan for features - representation



Variabile	Asse 1	Asse 2	Asse 3	Asse 4	Asse 5
cereali	-0,59	-0,36	-0,19	0,60	0,20
riso	-0,69	-0,27	0,14	-0,61	-0,07
patate	-0,21	-0,07	0,95	0,10	0,11
zucchero	0,87	0,01	-0,16	0,18	0,04
verdure	-0,89	0,05	-0,12	0,13	-0,28
vino	-0,73	0,35	-0,30	-0,17	0,34
carne	-0,24	0,81	0,32	0,25	0,15
latte	0,79	-0,39	0,16	-0,02	-0,02
burro	0,50	0,64	-0,09	-0,28	0,31
uova	0,07	0,86	0,01	0,09	-0,43

- Positive linear correlation when the angle between two vectors is nearly 0°
- Negative linear correlation when the angle between two vectors is 180°
- Uncorrelation when the angle between two vectors is 90°

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The Reduced-Rank Model definition

- SVD provides an optimal least squares approximation of the matrix $\mathbf{X} = [x_{ij}]$ of rank $r = \min(n, p)$ by another matrix $\widehat{\mathbf{X}} = [\hat{x}_{ij}]$ of reduced rank $\mathrm{M} < r$.
- The **reduced-rank model** is the systematic component of the statistical model identity:

$$x_{ij} = \sum_{m=1}^{M} \tau_m v_{im} u_{jm} + \varepsilon_{ij}$$

- Where $M < r = \min(n, p)$.
- The approximation is provided by the least squares fit:

$$\hat{x}_{ij} = \sum_{m=1}^{M} \tau_m v_{im} u_{jm}$$

- so that $x_{ij} \cong \hat{x}_{ij}$.
- PCA can be understood as a reduced-rank model.
- *Using this formulation, PCA can be applied iteratively for missing data imputation when data are missing at random.*

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The Percentage of Variance Explained by the PCs

- The Total Deviance of all features can be decomposed into the explained deviance due to the PCs and the Residual Sum of Squares as follows:

$$\sum_{j=1}^{p}\sum_{i=1}^{n} x_{ij}^2 = \sum_{m=1}^{M}\sum_{i=1}^{n} \hat{x}_{ij}^2 + \sum_{j=1}^{p}\sum_{i=1}^{n}\left(x_{ij} - \sum_{m=1}^{M} \tau_m v_{im} u_{jm}\right)^2$$

- The Percentage of Variance Explained (PVE) by the first M Principal Components is given by:

$$PVE = 1 - \frac{RSS}{TSS}$$

- We can interpret the PVE as the $R^2$of the approximation for the matrix $\mathbf{X}$ given by the first M principal components.

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Exact Biplot and Rank Two Matrix

(Gabriel, 1971; Gower and Hand, 1995)

- Let Z be the (n, p) matrix of standardized variables.
- Assume the reduced-rank of two dimensions: $h = 2 < r = \min(n, p)$ so that the eigenvectors matrices U and V have dimensions $(n, 2)$ and $(p, 2)$ respectively.
- The exact biplot can be derived as follows:

$$Z \cong V\Gamma U' = V\Gamma^{\delta}\Gamma^{1-\delta}U' = GH'$$

with $0 \leq \delta \leq 1$, $G = V\Gamma^{\delta}, H' = \Gamma^{1-\delta}\, U'$.

- In scalar form:

$$z_{ij} \cong g_i{}'h_j$$

where

✓ $g_i{}'$ is the i-th row 2-dimensional vector of G (the scores for the first two principal components);

✓ h_j is the j-th column 2-dimensional vector of H (the principal component loadings).

- For $\delta = 0$, g_i' are the normalized scores of the i-th unit on the first two Principal Components and h_j are the correlations of each Principal Component with the j-th variable.
- For $\delta = 1$, g_i' are the non normalized scores and h_j are the normalized eigenvectors.
- Typical choice is $\delta = 0.5$, i.e., a symmetrically scaled biplot with no distance interpretation to the rows or the columns, but only the scalar product interpretation.

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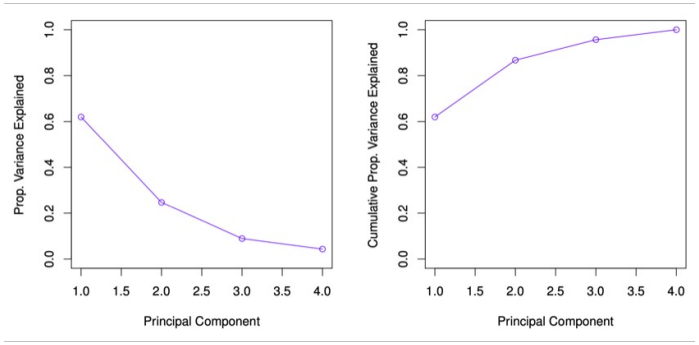
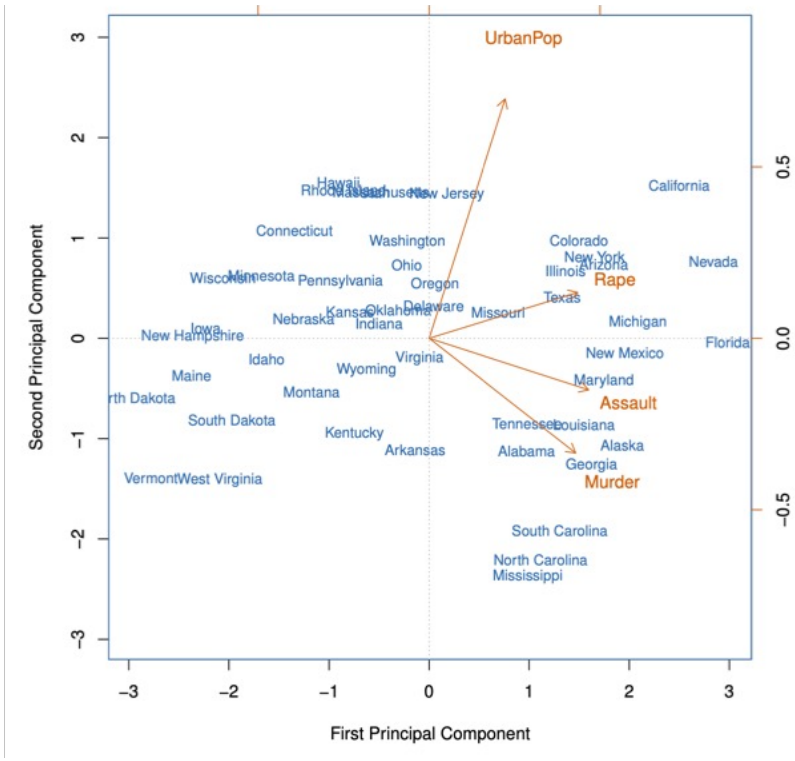
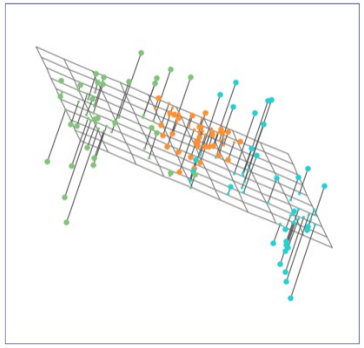
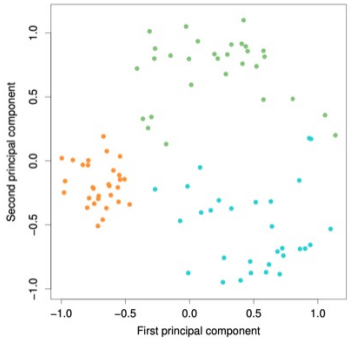
Example of biplot

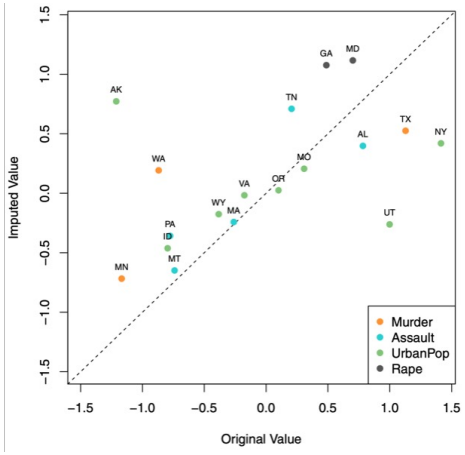
USArrests data *(Textbook "the Introduction to Statistical Learning")*

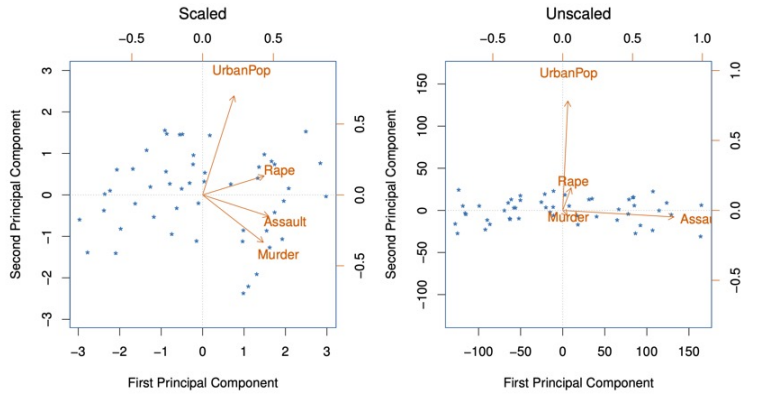


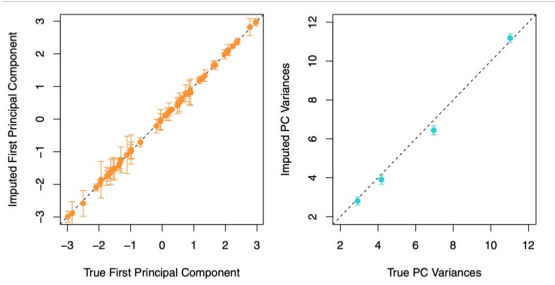












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Primer

Check for updates

Principal component analysis

Michael Greenacre[1], Patrick J. F. Groenen[2], Trevor Hastie[3], Alfonso Iodice D'Enza[4], Angelos Markos[5] & Elena Tuzhilina[6]

Abstract

Principal component analysis is a versatile statistical method for reducing a cases-by-variables data table to its essential features, called principal components. Principal components are a few linear combinations of the original variables that maximally explain the variance of all the variables. In the process, the method provides an approximation of the original data table using only these few major components. This Primer presents a comprehensive review of the method's definition and geometry, as well as the interpretation of its numerical and graphical results. The main graphical result is often in the form of a biplot, using the major components to map the cases and adding the original variables to support the distance interpretation of the cases' positions. Variants of the method are also treated, such as the analysis of grouped data, as well as the analysis of categorical data, known as correspondence analysis. Also described and illustrated are the latest innovative applications of principal component analysis: for estimating missing values in huge data matrices, sparse component estimation, and the analysis of images, shapes and functions. Supplementary material includes video animations and computer scripts in the R environment.

Sections

Introduction
Experimentation
Results
Applications
Reproducibility and data deposition
Limitations and optimizations
Outlook

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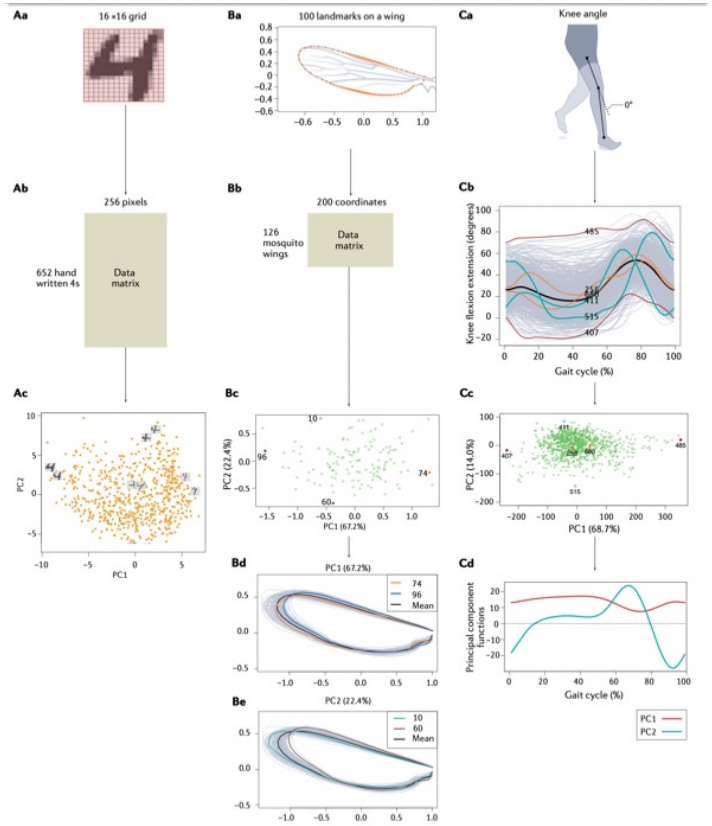

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Package	Function	Description
R packages		
stats	prcomp, princomp	These base R functions have minimal output and sometimes conflicting terminology.
base	svd	Singular value decomposition of a matrix.
FactoMineR[140]	PCA	The FactoMineR, ade4, amap, easyCODA and PCAtools packages all have options for weighting rows and columns of the data matrix. Options for supplementary rows and supplementary columns are provided in PCA (FactoMinerR) and dudi.pca (ade4). Note that PCA (easyCODA) has supplementary rows only. The FactoMineR, ade4 and easyCODA packages have extensive results in the created objects. The easyCODA package is aimed at compositional data analysis but has functions for PCA, correspondence analysis, LRA and RDA. Most of these packages have dedicated plotting functions (in the case of the ade4 package there is a separate package adegraphics[141]).
ade4[142]	dudi.pca	
amap	acp	
easyCODA[21]	PCA	
PCAtools	pca	
pca3d	pca3d	Three-dimensional PCA graphics.
vegan	rda	This function computes RDA, that is, PCA with constraints, but can also perform PCA with no constraints. The same package has function cca for correspondence analysis with or without constraints.
elasticnet	spca	Implementations of sparse PCA using a lasso penalized least-squares approach to obtain sparsity. arrayspc is specifically designed for the case $p \gg n$, such as microarrays.
	arrayspc	
irlba	prcomp_irlba	These fast and memory-efficient functions (prcomp_irlba, svds, rpca, batchpcs and i_pca) are used when the data are too large to fit into memory, or are arriving in streams.
RSpectra	svds	
rsvd[143]	rpca	
onlinePCA	batchpca	
idm[144]	i_pca	
symbolicDA	PCA.centers.SDA	PCA for interval-valued data.
RSDA	sym.pca	
fdapace	FDA	PCA of functional data, where data are sparse and longitudinal.
softImpute	softImpute	Imputation of missing values for PCA or matrix completion; can handle very large and sparse matrices.
missMDA[145]	imputePCA	Imputation of missing values for PCA.
Python libraries		
scikit-learn[146]	sklearn. decomposition. PCA	PCA, also with truncated SVD for large datasets.
	sklearn. decomposition. SparsePCA	Sparse PCA using the lasso penalty.
	sklearn. decomposition. IncrementalPCA	Computes solution by processing data in chunks, when dataset is too large to fit into memory.
NumPy[147]	linalg.svd	SVD of a matrix.

LRA, logratio analysis; PCA, principal component analysis; RDA, redundancy analysis; SVD, singular value decomposition.

Box 2

Iterative algorithm for PCA with missing values

Step 1: initialization for rank $r=0$

(a) Set $\mathbf{X}_0=\mathbf{0}$.

(b) Replace the missing values in $\mathbf{X}$ by the corresponding values in $\mathbf{X}_0$.

(c) Compute residual sum-of-squares (RSS) between completed $\mathbf{X}$ and $\mathbf{X}_0$ and denote it by RSS_0.

Step 2: find solutions for ranks $r=1, 2, \ldots, p$ in a sequential way

(a) Iterate the following steps until convergence:

(i) Compute the first r principal components of completed $\mathbf{X}$, obtaining the rank r approximation $\mathbf{X}_r$ from the singular value decomposition as follows:

$$\mathbf{X}_r = \sum_{k=1}^{r} \alpha_k \mathbf{u}_k \mathbf{v}_k^T$$

(ii) Replace the missing values in $\mathbf{X}$ by the corresponding values in $\mathbf{X}_r$.

(b) At convergence, compute RSS between completed $\mathbf{X}$ and $\mathbf{X}_r$ and denote it by RSS_r. The proportion of variance explained by component r can be measured by $(\mathrm{RSS}_{r-1}-\mathrm{RSS}_r)/\mathrm{RSS}_0$.

Step 3: the proportions of variances explained by each component define the scree plot

Use it to choose a rank r^* for the final solution. Return the sample principal coordinates $\alpha_k\mathbf{u}_k$ and the variable standard coordinates $\mathbf{v}_k$ for $k=1, 2, \ldots, r^*$ that form the decomposition of $\mathbf{X}_{r^*}$.

Notes

- Because of the pre-centring, steps 1(a) and (b) amount to imputation with column means of the observed data.
- When proceeding from rank r to $r+1$ in step 2, the completed data matrix $\mathbf{X}$ carries the filled-in values from $\mathbf{X}_r$.
- Measuring RSS between completed $\mathbf{X}$ and $\mathbf{X}_r$ is equivalent to measuring RSS using the observed values only.

PCA, principal component analysis.

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